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Sapindus mukorossi Gaertn. and its bioactive metabolite oleic acid impedes methicillin-resistant *Staphylococcus aureus* biofilm formation by down regulating adhesion genes expression

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A R T I C L E I N F O

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A B S T R A C T

Plants are boon to the mankind due to plenty of metabolites with medicinal values. Though plants have traditionally been used to treat various diseases, their biological values are not completely explored yet. *Sapindus mukorossi* is one such ethnobotanical plant identified for various biological activities. As biofilm formation and biofilm mediated drug resistance of methicillin-resistant *Staphylococcus aureus* (MRSA) have raised as serious global issue, search for antibiofilm agents has gained greater importance. Notably, antibiofilm potential of *S. mukorossi* is still unexplored. The aim of the study is to explore the effect of *S. mukorossi* methanolic extract (SMME) on MRSA biofilm formation and adhesive molecules production. Significantly, SMME exhibited 82 % of biofilm inhibition at 250 μg/mL without affecting the growth and microscopic analyses evidenced the concentration dependent antibiofilm activity of SMME. *In vitro* assays exhibited the reduction in slime, cell surface hydrophobicity, autoaggregation, extracellular polysaccharides substance and extracellular DNA synthesis upon SMME treatment. Further, qPCR analysis confirmed the ability of SMME to interfere with the expression of adhesion genes associated with biofilm formation such as *icaA*, *icaD*, *fnbA*, *fnbB*, *clfA*, *cna*, and *altA*. GC–MS analysis and molecular docking study revealed that oleic acid is responsible for the antibiofilm activity. FT-IR analysis validated the presence of oleic acid in SMME. These results suggest that SMME can be used as a promising therapeutic agent against MRSA biofilm-associated infections.

1. Introduction

Staphylococcus aureus is a Gram-positive bacterium which is an opportunistic human pathogen with high morbidity. *S. aureus* is known to cause soft tissue skin infections such as pimples, boils, impetigo, abscesses, carbuncles and also life-threatening conditions such as bacteremia, septicemia, pneumonia and surgical site infections (Troeman et al., 2019). Formation of biofilm under *in vivo* condition leads to persistent chronic infection which is unresponsive to all the available treatment options and makes the condition fatal (Tong et al., 2015). Biofilm is the hydrated mat like structure of microbial cells living together and provides the protective room for the bacterial cells from host immune response and antimicrobial treatments. Conventional antimicrobial treatment strategy has become ineffective due to the emergence of multi drug resistant strains of bacteria with biofilm formation (Verderosa et al., 2019). Methicillin-resistant *Staphylococcus aureus* (MRSA) is one such strain which is the leading cause of hospital associated infections in humans (Turner et al., 2019). In 2017, world health organization (WHO) categorised MRSA as high priority multi drug resistant pathogen and this alarms the prevalence of MRSA and need of alternative therapeutic strategies to fight persistent infections of MRSA (WHO, 2017). MRSA produces an array of adhesive molecules to

Abbreviation: ATCC, American type culture collection; CLSM, Confocal laser scanning microscopy; CRA, Congo red agar; eDNA, Extracellular DNA; EPS, Extracellular polymeric substance; FTIR, Fourier-transform infrared spectroscopy; GC–MS, Gas chromatography; HI, Hydrophobicity index; MATH, Microbial adhesion to hydrocarbon; MBIC, Minimum biofilm inhibitory concentration; MRSA, Methicillin-resistant*Staphylococcus aureus*; qPCR, Real-Time quantitative PCR; SD, Standard deviation; SMME, *Sapindus mukorossi methanolic extract*; WHO, World health organization.

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colonize the host tissue and these specialized molecules having the ability to interact with host extra cellular matrix are called as microbial surface component recognizing adhesive matrix molecules (MSCRAMMs). This group comprises of fibronectin, fibrinogen, elastin and collagen-binding proteins. Impairment of production of these proteins will inhibit the surface adherence of MRSA thereby makes the cells free floating which can be easily encountered either by immune system or by antimicrobial treatment (Dakheel et al., 2016; Madani et al., 2017). Henceforth, identification of antibiofilm agents from natural resources is the recent strategy to potentiate antibiotic therapy in the treatment of chronic microbial conditions.

Sapindus mukorossi Gaertn., a member of the family *Sapindaceae*, is an ethnobotanically important tree of tropical and sub-tropical regions of Asia and particularly found in Gangetic Plains, Western Ghats, and Deccan Plateau of India. It is commonly called by several names includes soapnut, soapberry, washnut, reetha, aritha, dodan and doadni. Though more than 40 wild species have been identified in the genus Sapindus, especially *S. mukorossi* has got attracted by researchers due to its numerous biological and pharmacological properties (Chen et al., 2019). The dried fruit of *S. mukorossi* has a soapy texture because of the presence of saponins and has traditionally been used to prepare Ayurvedic shampoos, detergents and hand wash for its biosurfactant activity. In addition, *S. mukorossi* is reported to have various pharmacological properties such as anticancer, anti-inflammatory, antiparasitic, antiviral, antifungal and antibacterial activities (Shah et al., 2017). Especially, potential of *S. mukorossi* seeds in the treatment of microbial infections and skin care was described 500 years ago in Compendium of Materia Medica (Bencao Gangmu in Mandarin), a traditional pharmaceutical book from China (Chen et al., 2019). Traditionally, powdered seed of *S. mukorossi* was utilized in the treatment of dental carries, common cold and nausea (Francis et al., 2002; Shah et al., 2017). Several studies also reported the antibacterial property of *S. mukorossi* against clinically important bacteria (Sharma et al., 2013; Heng et al., 2014; Wei et al., 2020). Nevertheless, no report is available on antibiofilm efficacy of *S. mukorossi* and the present study is the first ever report to study the antibiofilm potential of *S. mukorossi* against serious human pathogen MRSA.

2. Materials and methods

2.1. Bacterial strain and culture conditions

MRSA (ATCC-33591) was obtained from the ATCC, HiMedia, India. It was grown on tryptone soya broth/agar (TSB/TSA) at 37 °C and maintained at 4 °C for future use. For biofilm assays, 1% sucrose was added along with TSB medium (TSBS).

2.2. Plant material

S. mukorossi (Vernacular name: Boondi kottai and Reetha) seeds were purchased from local market of Karaikudi, Tamil Nadu, India. Then, the plant specimen was deposited and identified as *Sapindus mukorossi* Gaertn. (Voucher No. ASR001) by Dr. John Britto, Director, The Rapinat Herbarium and Centre for Molecular Systematics, St. Joseph's College, Tiruchchirappalli, Tamil Nadu, India. Further, we declared that any endangered or protected species was not used this study.

2.3. S. mukorossi methanolic extract (SMME) preparation

The seed pericarps were powdered using mixer grinder and then seed pericarp powder (10 g) was mixed with 100 mL of methanol at 37 °C for 24 h in orbital shaker (160 rpm). After 24 h, the extract was filtrated by Whatman filter paper and the filtrate was concentrated by converting into powder form using rotary evaporator at 50 °C. Further, 10 mg/mL of SMME was prepared as stock solution in methanol and used for further experiments (Srinivasan et al., 2016).

2.4. Antibiofilm potential of SMME

2.4.1. Determination of minimum biofilm inhibitory concentration (MBIC) of SMME

In 24-well plate, 10 μL of MRSA overnight culture was inoculated into 1 mL of TSBS with increasing concentrations of SMME (62.5, 125, 250 and 500 μg/mL) and incubated for 24 h at 37 °C. A 25 μL of methanol was added to 1 mL of TSBS media containing 10 μL of MRSA culture and it was considered as vehicle control to assess the influence of solute (methanol) on MRSA biofilm. Wells containing TSBS alone and wells with 300 μg/mL of eugenol (Alfa Aesar, Thermo Fisher Scientific, India) were considered as blank and positive control respectively. After incubation, the culture plate was observed at 600 nm (Spectramax M3, Molecular Devices, United States) to determine cell density of control and treated samples. For biofilm quantification, planktonic cells were discarded from 24-well plate and the wells with biofilm were washed thrice with phosphate-buffered saline (PBS) and air dried. Then, 0.4 % crystal violet solution was used to stain the biofilm and excess stain was removed by PBS wash and 24-well plate was imaged. Subsequently, 30 % of glacial acetic acid was used to dissolve crystal violet and the titre plate was read at 570 nm. The biofilm inhibition was calculated as percentage by the following formula: Biofilm inhibition (%) = [(Control OD_{570nm} – Treated OD_{570nm})/Control OD_{570nm}] × 100 (Chatterjee et al., 2020).

2.4.2. Ring biofilm assay

For ring biofilm formation, 2 mL of TSBS in glass test tubes with 20 μL of overnight culture of MRSA was treated with increasing concentrations of SMME (62.5, 125, 250 and 500 μg/mL) and incubated for 48 h at 37 °C. Parallelly, blank and positive control (300 μg/mL of eugenol) was maintained. Afterward, the planktonic cells were discarded and washed with PBS and air-dried. Then, the tubes were stained with 0.4 % crystal violet solution. Air- liquid interface ring biofilm formation was visually observed and photographically imaged (Sethupathy et al., 2017).

2.4.3. Microscopic analyses

The antibiofilm activity of SMME was validated through light and confocal laser scanning microscopy (CLSM). The biofilm assay was performed on a 24-well plate containing 1 × 1 cm glass slides. The planktonic cells of control, positive control (300 μg/mL of eugenol) and SMME treated (62.5, 125 and 250 μg/mL) slides were removed by washing with PBS and air-dried. Then, the glass slides were stained with 0.4 % crystal violet solution for light microscope and 0.2 % acridine orange stain used for CLSM analysis. After staining, glass slides were observed under the light microscope (Nikon Eclipse Ti 100, Tokyo, Japan) at 400X magnification and imaged. For CLSM, the glass slides were imaged at 200X under CLSM microscope (Model LSM 710, Carl Zeiss, Oberkochen, Germany) (Prasath et al., 2019).

2.5. Influence of SMME on MRSA growth

2.5.1. Growth curve analysis

The growth curve assay was performed to determine the effect of SMME on growth pattern of MRSA. One percent of MRSA culture was used to inoculate the 100 mL of TSBS with and without SMME (250 μg/mL) and incubated for 24 h at 37 °C. For every 1 h, the cell density of control and SMME treated samples were read at 600 nm up to 24 h (Sethupathy et al., 2017).

2.5.2. Resazurin cell viability assay

The cell viability of MRSA control and SMME treated samples were assessed by resazurin cell viability assay. The resazurin solution was freshly prepared as a stock solution (6.5 mg/mL) in PBS. For cell viability assay, both the MRSA control and SMME treated cells were grown for 24 h at 37 °C. Then, the cell pellets were separated by centrifugation at 8000 rpm for 10 min and washed thrice with PBS and resuspended in PBS. A 20 μL of resazurin was added to each well of a 96-well microtiter plate containing 180 μL of cell suspensions of control and SMME treated samples and incubated in the dark for 4 h at 37 °C. After incubation, supernatants of the samples were separated by centrifugation at 12,000 rpm for 10 min. Fluorescence intensity of collected supernatants was measured using spectrophotometer at 590 nm emission and 560 nm excitation wavelengths (Selvaraj et al., 2019).

2.6. *Influence of SMME on adhesion factors*

2.6.1. *Qualitative analysis of slime synthesis*

Slime synthesis was assessed by Congo red agar (CRA) plate (TSB-30 g/L, Agar-1.8 g/L and Congo red dye – 0.8 g/L) method to determine the influence of SMME on slime production by MRSA. The CRA plates were prepared with and without SMME at increasing concentrations (62.5, 125 and 250 μg/mL). Overnight MRSA culture was streaked on CRA plates and incubated at 37 °C for 24 h. Plates were imaged after incubation period (Kannappan et al., 2019).

2.6.2. *Microbial adhesion to hydrocarbon (MATH) assay*

MATH assay was performed to assess the effect of SMME on the MRSA cell surface hydrophobicity (CSH). MRSA was grown on test tubes containing 1 mL of TSBS in the absence and presence of SMME (62.5, 125 and 250 μg/mL) for 24 h at 37 °C. After the period of incubation, the cells were collected by centrifugation at 8000 rpm for 10 min and washed twice with PBS. Again, the pellets were resuspended in 1 mL of PBS and OD at 600 nm was adjusted to 0.9 OD ± 0.05. Then, the cell suspension was transferred to test tubes followed by the addition of an equal volume of toluene, vortexed vigorously for 2 min and incubated at cold temperature for 24 h. After incubation, the aqueous phase was collected to measure OD at 600 nm and the percentage of hydrophobicity index (HI) was calculated using the formula: HI (%) = [1-

Table 1
List of primers used for qPCR analysis.

Genes	Forward primer	Reverse primer
gyrB	5′-GGTGCTGGGCAAATACAAGT-3′	5′-TCCCACACTAAATGGTGCAA-3′
icaA	5′-ACACTTGCTGGCGCAGTCAA-3′	5′-TCTGGAACCAACATCCAACA-3
icaD	5′-ATGGTCAAGCCCAGACAGAG-3′	5′-AGTATTTTCAATGTTTAAAGCA-3′
fnbA	5′-ATCAGCAGATGTAGCGGAAG-3′	5′-TTTAGTACCGCTCGTTGTCC-3′
fnbB	5′-AAGAAGCACCGAAAACTGTG-3′	5′-TCTCTGCAACTGCTGTAACG-3′
cna	5′-AAAGCGTTGCCTAGTGGAGA -3′	5′-AGTGCCTTCCCAAACCTTT-3′
altA	5′-TGTCGAAGTATTTGCCGACTTCGC-3′	5′-TGGAATCCTGCACATCCAGGAAC-3′

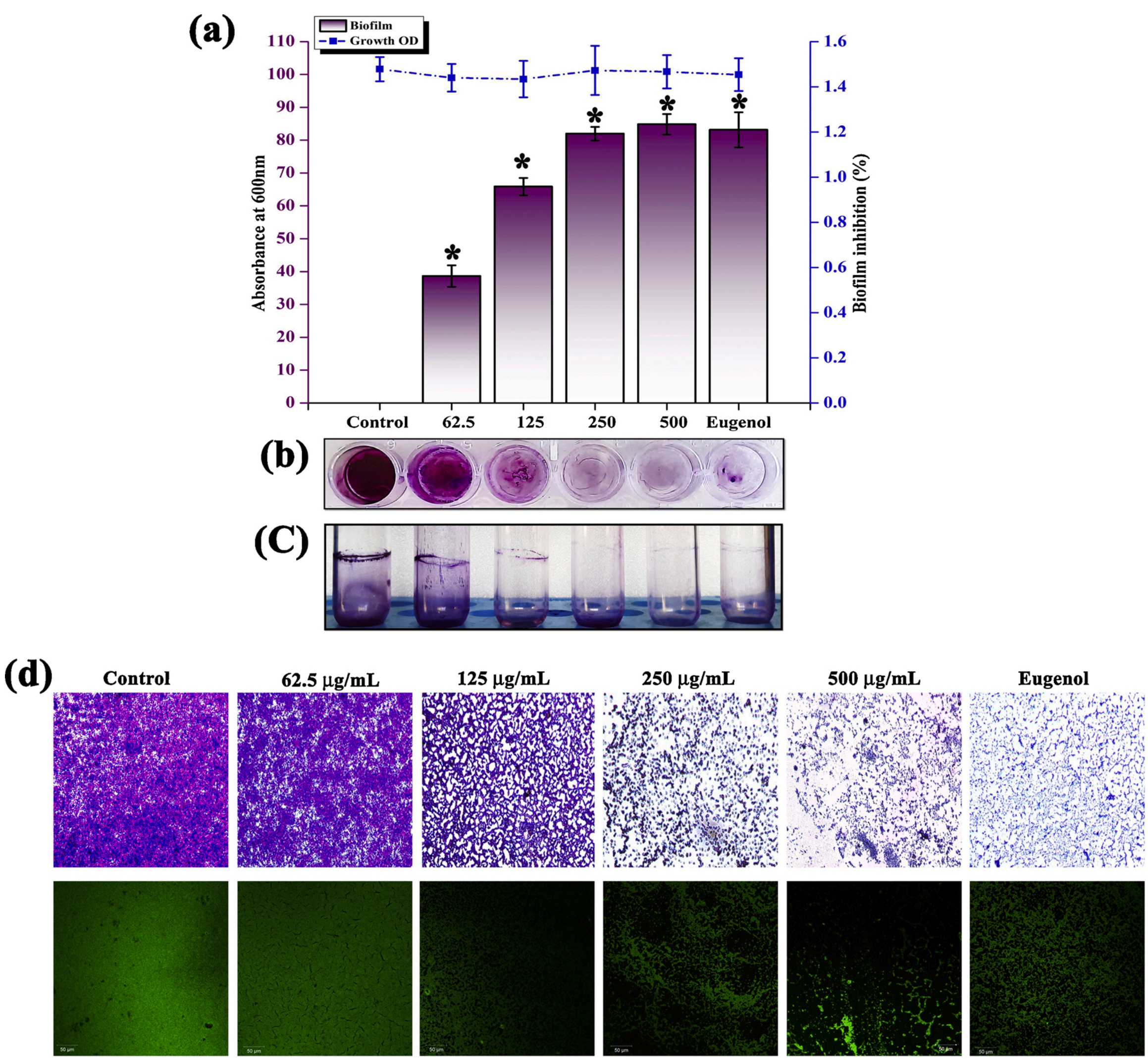

Fig. 1. Concentration dependent antibiofilm activity of SMME against MRSA. (a) Bar graph indicates biofilm inhibition and line graph indicates growth. Effect of SMME on MRSA biofilm formation on glass (b) and polystyrene surfaces (c). Light and CLSM micrographs showing the antibiofilm potential of SMME against MRSA (d). Error bars indicate standard deviations and asterisks indicate the statistical significance ($p < 0.05$).

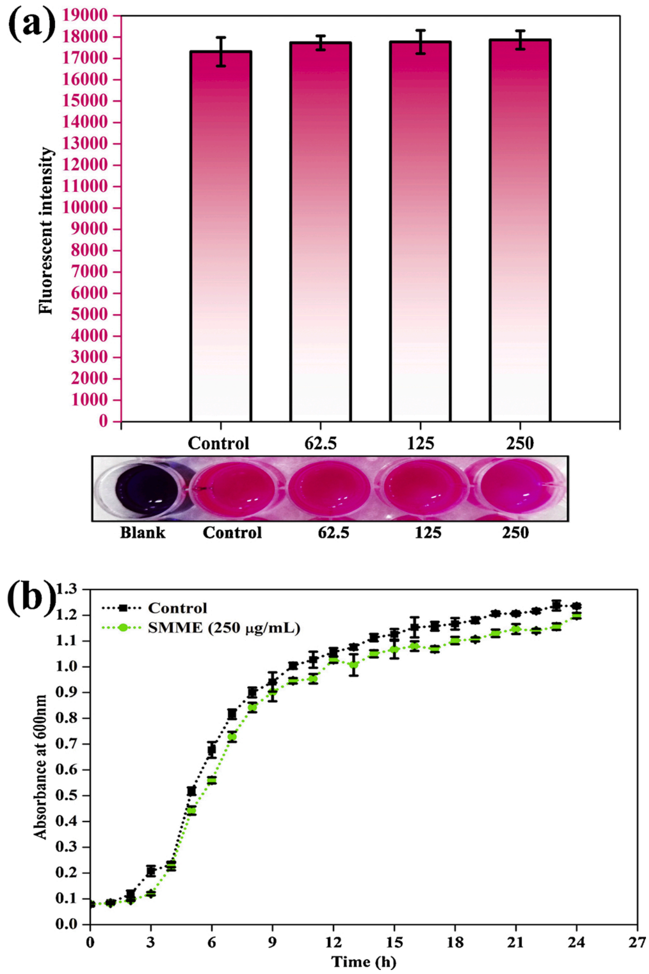

Fig. 2. Assessment of the effect of SMME (62.5, 125 and 250 µg/mL) on viability of MRSA using resazurin assay (a). Growth curve of MRSA in the presence and absence of SMME at 250 µg/mL concentration (b). Error bars indicate standard deviations.

(OD_{600nm} after vortex/OD_{600nm} before vortex)] X 100 (Sivasankar et al., 2016).

2.6.3. Extracellular polymeric substances (EPS) quantification assay

The phenol-sulfuric acid method was used to estimate and detect the impact of SMME treatment on the EPS of MRSA. To quantify EPS, MRSA was grown with and without SMME at increasing concentrations (62.5, 125 and 250 µg/mL) on a 6-well plate for 24 h at 37 °C. After incubation, the planktonic cells were discarded without disturbing the biofilms and 500 µL of TE buffer (10 mM EDTA in 10 mM Tris buffer) was used to scrape the biofilm from the wells of the plate. To the biofilm mixture, 500 µL of 5% phenol and 5 mL sulfuric acid were added and incubated in dark chamber for 1 h followed by the measurement of absorbance at 490 nm (Kannappan et al., 2019).

2.6.4. Autoaggregation assay

MRSA was grown in test tubes containing 10 mL of TSBS with and without SMME (62.5, 125 and 250 µg/mL) for 24 h at 37 °C. After incubation, the cell pellets were collected by centrifugation at 8000 rpm for 10 min and washed thrice with PBS. Again, the pellet was resuspended in 10 mL of PBS and kept undisturbed for 24 h. A 100 µL of cell suspensions from the top portion of test tubes were collected for every 3 h and cell density was measured by OD at 600 nm up to 24 h. The result was visually observed and imaged (Sethupathy et al., 2017).

2.6.5. Autolysis assay

The effect of SMME on autolysis of MRSA was analysed by collecting pellets from both MRSA control and SMME (62.5, 125 and 250 µg/mL) treated samples. The collected cell pellets were washed twice with ice-cold PBS, resuspended in autolysis buffer (0.05 % Triton X-100 and 0.05 M Tris-HCL in PBS) and incubated at 30 °C. OD at 600 nm was measured for every 30 min up to 3 h (Sethupathy et al., 2017).

2.6.6. Extracellular DNA (eDNA) quantification

MRSA was grown on 6-well plate containing 5 mL of TSBS with and without SMME (62.5, 125 and 250 µg/mL) and incubated at 37 °C for 24 h. After incubation, the planktonic cells were carefully removed and plate wells were washed with PBS to remove unbound cells. One millilitre of TE buffer was added to remove adhered biofilm cells. Then, the solution with biofilm cells was transferred to 1.5 mL tubes and centrifuged at 8000 rpm for 10 min to remove supernatant. To the pellet, 200 µL of TE buffer was added and vortexed vigorously. Supernatant was collected by centrifugation at 12,000 rpm for 15 min. The amount of eDNA present in each sample was quantified using nano-spectrometer. Then, the samples were qualitatively analyzed by agarose gel (1.5 %) electrophoresis (Valliammai et al., 2019).

2.7. Gene expression analysis by quantitative PCR (qPCR)

Total RNA was isolated from control and SMME (250 µg/mL) treated samples by the Trizol method and cDNA was synthesized using High Capacity cDNA Reverse Transcription Kit (Applied Biosystems, USA) by the manufacturers' protocol. Then, the expression of adhesion-associated genes including *icaA*, *icaD*, *fnbA*, *fnbB*, *clfA*, *cna*, and *altA* was examined by qPCR analysis using PCR mix (SYBR Green kit, Applied Biosystems, United States) at predefined ratio. *gyrB* housekeeping gene was used to normalize the cycle threshold (Ct) values of all the tested genes and $2^{-\Delta\Delta Ct}$ method was used to calculate difference in gene expression (Valliammai et al., 2019). The details of the primer sequences of tested genes are given in Table 1.

2.8. Identification of active principles of SMME

2.8.1. Successive solvent extraction

The powder of *S. mukorossi* pericarps was taken for polar to non-polar solvent extraction. SMME was subjected to successive solvent extraction by following the previously reported protocol with slight modification (Muthamil et al., 2018). Briefly, 10 g of this powder was extracted using 100 mL of various solvents in the following order Milli-Q water > dimethyl sulfoxide (DMSO) > methanol > ethyl acetate > chloroform > petroleum ether > n-hexane and then filtered by Whatman filter paper. Then, solvent filtrates were evaporated at 50 °C and vacuum dried. Further, extracted powders were prepared as 10 mg/mL stock in methanol and subjected to biofilm assays as described earlier.

2.8.2. Gas chromatography-mass spectrometry (GC–MS) analysis and evaluation of antibiofilm potential of bioactive lead

In order to identify the active principle responsible for antibiofilm activity of SMME, methanolic fraction was subjected to the GC–MS analysis. GC–MS analysis was performed using Agilent 7000D Triple Quadrupole GC/MS system with HP-5 ms ultra inert column (length 30 m and diameter 0.25 mm) and helium as a carrier gas with constant flow rate of 1 mL per min. Then, 1 µL of sample was injected into the system at an injection temperature of 250 °C. Initially, the oven temperature was kept at 50 °C for 2 min and increased to 300 °C at the rate of 10 °C/min. For mass detector, the transfer line temperature and ion source temperature were 300 °C and 280 °C respectively. The ionization mode electron impact was set as 70 eV with 0.2 s of scan time and 0.1 s of scan interval and from 45 to 600 Da of fragments. The spectrum of compounds from SMME was compared with the known compounds spectrum of database library (GC–MS NIST 2017) (Muthamil et al., 2018;

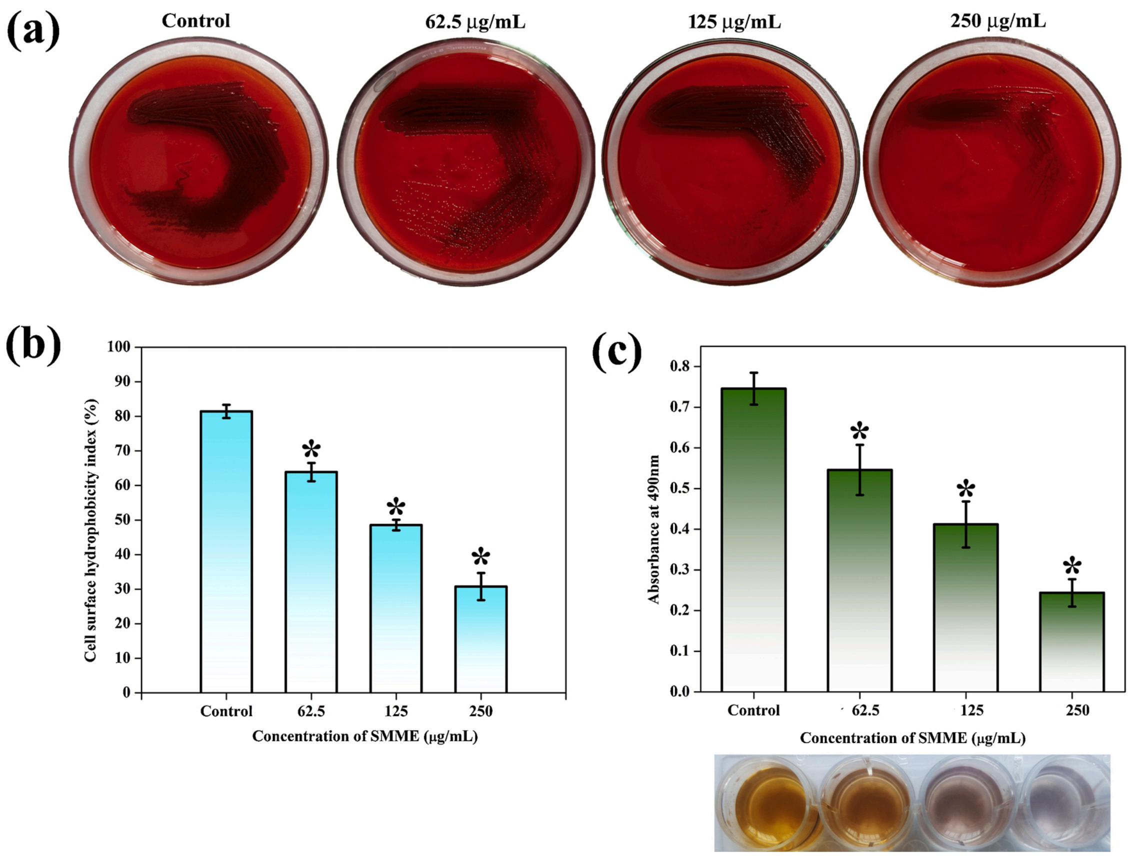

Fig. 3. Inhibitory potential of SMME at increasing concentrations (62.5, 125 and 250 μg/mL) on slime synthesis (a), cell surface hydrophobicity (b) and extracellular polymeric substances (c) of MRSA. Error bars indicate standard deviations and asterisks indicate the statistical significance ($p < 0.05$).

Subramenium et al., 2018).

2.8.3. *Molecular docking study*

To determine the bioactive active compounds of SMME responsible for antibiofilm activity against MRSA, molecular docking studies were done with SarA protein which has been reported as major regulator associated with biofilm formation in MRSA. Therefore, SarA (ID: 2FNP) crystal structure was retrieved from Protein Data Bank (PDB). Then, the three-dimensional structures of bioactive compounds of SMME were obtained from PubChem database. Autodock Tools *v*1.5.6 was used to assess the interactions between bioactive compounds of SMME and SarA of MRSA. Furthermore, BIOVIA Discovery Studio visualizer 2016 *v*16.1.0.15350 was used to visualize docked complexes (Norgan et al., 2011; Chemmugil et al., 2019).

2.8.4. *Fourier-transform infrared spectroscopy (FTIR) analysis*

FTIR (Nicolet™ iS™ 5 FTIR Spectrometer, Thermo Scientific, USA) analysis was performed to compare the similarity between the SMME and standard oleic acid (Alfa Aesar, Thermo Fisher Scientific, India). An equal volume of each samples were placed at attenuated total reflectance (ATR) plate. FTIR spectra of SMME and oleic acid were collected at the frequency range of 4000−400 cm^{-1} using the OMNIC software (Subramenium et al., 2018).

2.9. *Statistical analysis*

All the experiments were performed in three biological replicates with the minimum of two technical replicates and all the data were presented as mean ± standard deviation (SD). Statistical significance was assessed by one-way ANOVA method followed by Duncan's post hoc test using SPSS 17.0 software package (SPSS Inc., Chicago, IL) and *p*-value < 0.05 was fixed as significant.

3. Results and discussion

3.1. *Antibiofilm and non-antibacterial potential of SMME against MRSA*

S. aureus is a commensal bacterium and it becomes a notorious pathogen due to its biofilm forming ability. Biofilm of *S. aureus* is well-known as an essential factor for its antibiotic resistance, persistence and pathogenesis. Biofilm formed by MRSA on indwelling medical devices are very difficult to eradicate and it leads to serious infections in the health care setting (Archer et al., 2011). In the previous study, among the 54 isolates of *S. aureus* tested for biofilm formation and antibiotic susceptibility, 2 strains exhibited strong biofilm forming ability were found to have multi-drug resistance when compared to other 52 strains sensitive to the tested antibiotics namely penicillin, erythromycin, clindamycin and tetracycline (Lin et al., 2019). In addition, several studies have demonstrated that biofilm forming ability of *S. aureus* decreases its sensitivity to conventional antibiotics (Abdelhady et al., 2013; Algburi et al., 2017; Jimi et al., 2017; Craft et al., 2019). Further, MRSA was included in the list of high priority pathogens for the novel drug development by WHO (WHO, 2017). On the whole, biofilm forming ability of *S. aureus* play vital role in antibiotic resistance. Therefore, biofilm inhibition has been identified as an alternative therapeutic strategy to eliminate the biofilm-associated infections of MRSA. In the present study, antibiofilm potential of SMME was assessed against MRSA biofilm. Upon testing with the increasing concentrations of SMME on biofilm formation of MRSA, 250 μg/mL of SMME exhibited a maximum of 82 % biofilm inhibition without affecting growth and above this concentration percentage of biofilm inhibition was not considerably increased (85 % biofilm inhibition at 500 μg/mL). Hence,

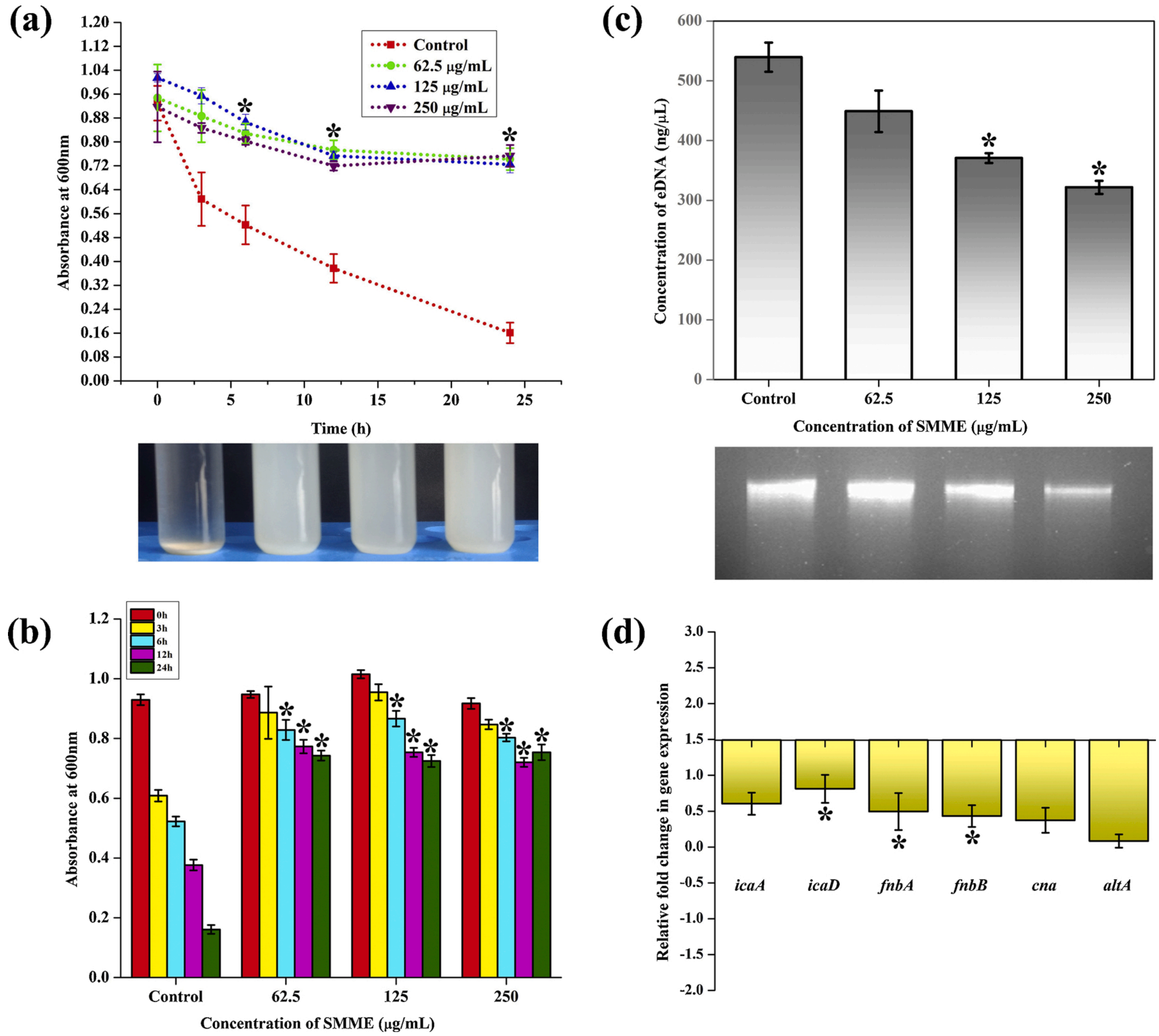

Fig. 4. Effect of SMME on autoaggregation (a), autolysis (b) and eDNA synthesis (c) in MRSA. qPCR analysis of expression pattern of genes involved in adhesion and biofilm formation of MRSA upon SMME treatment (d). Error bars indicate standard deviations and asterisks indicate the statistical significance ($p < 0.05$).

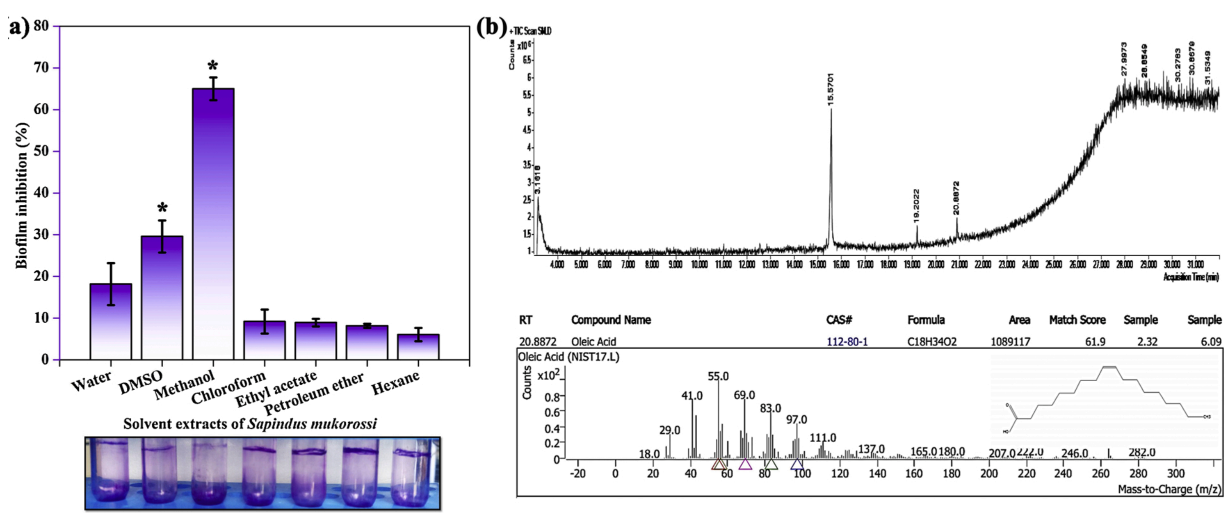

Fig. 5. Antibiofilm activity of different solvent extracts of *S. mukorossi* (a) and GC–MS analysis of methanolic extract of *S. mukorossi* (b). Error bars indicate standard deviations and asterisk indicates the statistical significance ($p < 0.05$).

250 μg/mL of SMME was set as MBIC (Fig. 1a). The antibiofilm potential of SMME was highly superior to previously reported gallic acid (Liu et al., 2017), biosurfactant of *Lactobacillus casei* (Merghni et al., 2017), alpha-mangostin extracted from *Garcinia mangostana* L. (Phuong et al.,

Table 2
List of compounds identified by GC–MS analysis.

S.No.	Compound Name	Area	Retention time
1	Ethyl iso-allocholate	17,143,476	3.1618
2	Octadecanal, 2-bromo-	17,143,476	3.1618
3	Hexadecanoic acid	17,143,476	3.1618
4	2-Myristynoyl pantetheine	17,143,476	3.1618
5	Scyllo-Inositol, 1-C-methyl-	17,886,933	15.5701
6	alpha-D-Mannofuranoside, 1-O-decyl-	17,886,933	15.5701
7	Myo-Inositol, 4-C-methyl-	17,886,933	15.5701
8	alpha-Methyl mannofuranoside	17,886,933	15.5701
9	Myo-Inositol, 2-C-methyl-	17,886,933	15.5701
10	Estra-1,3,5(10)-trien-17.beta.-ol	1,035,207	19.2022
11	Octadecanedioic acid	1,035,207	19.2022
12	Dasycarpidan-1-methanol	1,035,207	19.2022
13	Octadecanal	1,089,117	20.8872
14	Oleic Acid	1,089,117	20.8872
15	cis-Vaccenic acid	1,089,117	20.8872
16	cis-10-Heptadecenoic acid	1,089,117	20.8872
17	cis-13-Octadecenoic acid	1,089,117	20.8872
18	Demecolcine	568,992	27.7472

Table 3
Molecular docking analysis of bioactive compounds identified through GC–MS analysis with SarA receptor of MRSA.

S. No.	Compound (Ligand)	No of Hydrogen bonds	Key residue (s)	Binding energy (kcal/mol)
1	Ethyl iso-allocholate	1	Lys B: 127	−2.9
2	Octadecanal, 2-bromo-	1	Lys B: 123	−2.7
3	Hexadecanoic acid	1	Lys B: 127	−2.3
4	2-Myristynoyl pantetheine	2	Lys B: 127 Glu B: 135	−3.58
5	Scyllo-Inositol, 1-C-methyl-	1	Gln B: 166 Asn B: 161	−3.2
6	alpha-D-Mannofuranoside, 1-O-decyl-	3	Asp A: 120 Tyr B: 162	−3.67
7	Myo-Inositol, 4-C-methyl-	2	Asn B: 161 Gln B: 166	−3.6
8	alpha-Methyl mannofuranoside	2	Asp A: 120 Gln B: 166	−2.8
9	Myo-Inositol, 2-C-methyl-	2	Glu B: 135 Lys B: 123	−3.4
10	Estra-1,3,5(10)-trien-17. beta.-ol	3	Lys B: 123 Asp A: 120	−3.5
11	Octadecanedioic acid	2	Lys A: 127 Lys B: 127	−2.8
12	Dasycarpidan-1-methanol	1	Lys B: 123 Asp A: 120	−3.3
13	Octadecanal	1	Phe B: 134	−2.6
14	Oleic Acid	2	Glu B: 135 Lys B: 127	−6.85
15	cis-Vaccenic acid	–	–	−2.4
16	cis-10-Heptadecenoic acid	–	–	−3.6
17	cis-13-Octadecenoic acid	1	Lys B: 127	−2.3
18	Demecolcine	2	Lys B: 123 Asp A: 120	−3.0

2017) with 37, 70 and 40 % of biofilm inhibition respectively. Previous studies stated that the biofilm formed by MRSA on biotic/abiotic surfaces is a challenging medical condition as biofilm cells are much more resistant to antibiotics than free floating bacterial cells (Bhattacharya et al., 2015; Khatoon et al., 2018). Hence, in the present study the effect of SMME on adherence of MRSA on polystyrene and glass surface was assessed and SMME was found to be efficient in preventing the surface adherence of MRSA irrespective of surface nature (Fig. 1b and c). Light and CLSM microscopic analyses further validated the dose dependent inhibitory efficacy of SMME on surface adherence of MRSA (Fig. 1d). Particularly, 250 μg/mL of SMME greatly reduced the surface coverage when compared to the completely covered biofilm of untreated sample. Apart from inhibiting the adherence of MRSA on polystyrene surface, SMME was also able to reduce the ring biofilm formation of MRSA at air-liquid interface (Fig. 1b and c). Eugenol, an aromatic essential oil reported for antibiofilm activity against MRSA (Yadav et al., 2015; Al-Shabib et al., 2017) was included as positive control in the biofilm experiments and it was observed that the antibiofilm activity of SMME was equivalent to that of positive control eugenol (Fig. 1a–d). Non-antibacterial activity of antibiofilm agent is more advantageous as it does not exert any selection pressure on bacteria and thus excludes the possibility of resistance development. Growth curve of MRSA was unaffected in the presence of SMME and similarly, in resazurin assay fluorescence intensity of SMME treated cells was comparable with that of control cells (Fig. 2a and b). Both growth curve analysis and cell viability assay demonstrated that SMME does not interfere with the proliferation and metabolic activity of MRSA and confirming the non-antibacterial nature of SMME.

3.2. SMME inhibits adhesive molecules in MRSA

In the structure of biofilm, bacterial cells are embedded in the self-produced extracellular slime like matrix which is composed of numerous polymeric substances such as polysaccharides, DNA, proteins, lipids *etc.* This matrix like structure is otherwise called as slime because of its hydrated and sticky nature which provides the resilience and stability to biofilm (Lee et al., 2016). As SMME efficiently inhibited the biofilm, its effect on slime synthesis was examined. Results of CRA assay displayed the great reduction in slime synthesis upon SMME treatment (Fig. 3a). This result evidenced that synthesis of adhesive molecules essential for the biofilm formation is affected by SMME. Thus, the biofilm matrix formed in the absence and presence of SMME was collected and examined for the presence of EPS by phenol-sulfuric acid method. Results displayed the concentration dependent reduction in EPS upon SMME treatment (Fig. 3c). Cell surface hydrophobicity is another positive determinant which enhances the adhesion of bacteria to hydrophobic surfaces by increasing hydrophobic interactions (Sivasankar et al., 2016). MATH assay exhibited that hydrophobicity index was reduced up to 50 % in the presence of SMME (Fig. 3b). Reduction of hydrophobicity by SMME could be responsible for reduction in adherence to hydrophobic polystyrene surfaces. Another interesting factor is that SMME treated MRSA cells were always appeared to be dispersed while comparing to aggregated control cells. This fact has driven the study to investigate the pattern of autoaggregation of MRSA for 24 h in the presence of SMME by test tube settling assay. Results unraveled that SMME impaired the aggregating ability of MRSA as SMME treated cells were found to be unsettled down even after 24 h whereas most of the control cells were settled down within 12 h (Fig. 4a). Autoaggregation is a crucial property of MRSA which is essential for intracellular adherence and surface adherence as well. Particularly, intracellular adhesion in MRSA is mediated by polysaccharide intracellular adhesin (PIA) encoded by *icaABCD* operon (Arciola et al., 2015). In addition, few studies reported that intracellular adhesion is also mediated by eDNA which is released by the autolysis of the few populations of bacterial cells resides in the biofilm. eDNA is reported to provide the structural stability to biofilm by means of intracellular adhesion (Sugimoto et al., 2018). Examination of autolysis and eDNA synthesis in the presence of SMME demonstrated that SMME treated cells were resistant to triton-X induced autolysis whereas controls cells were susceptible to triton-X treatment and lysed faster (Fig. 4b). Quantification of eDNA demonstrated the considerable reduction in eDNA (321 ng/μL) upon SMME treatment when compared to the amount of eDNA (539 ng/μL) in control sample (Fig. 4c). This result is in correlation with the autolysis assay as autolysis is responsible for the release of more nucleic material into matrix. Further, in order to find out the effect of SMME on virulence gene expression, qPCR analysis was performed. Results revealed that SMME treatment reduced the expression of *icaA*, *icaD*, *fnbA*, *fnbB*, *cna* and *atlA*.

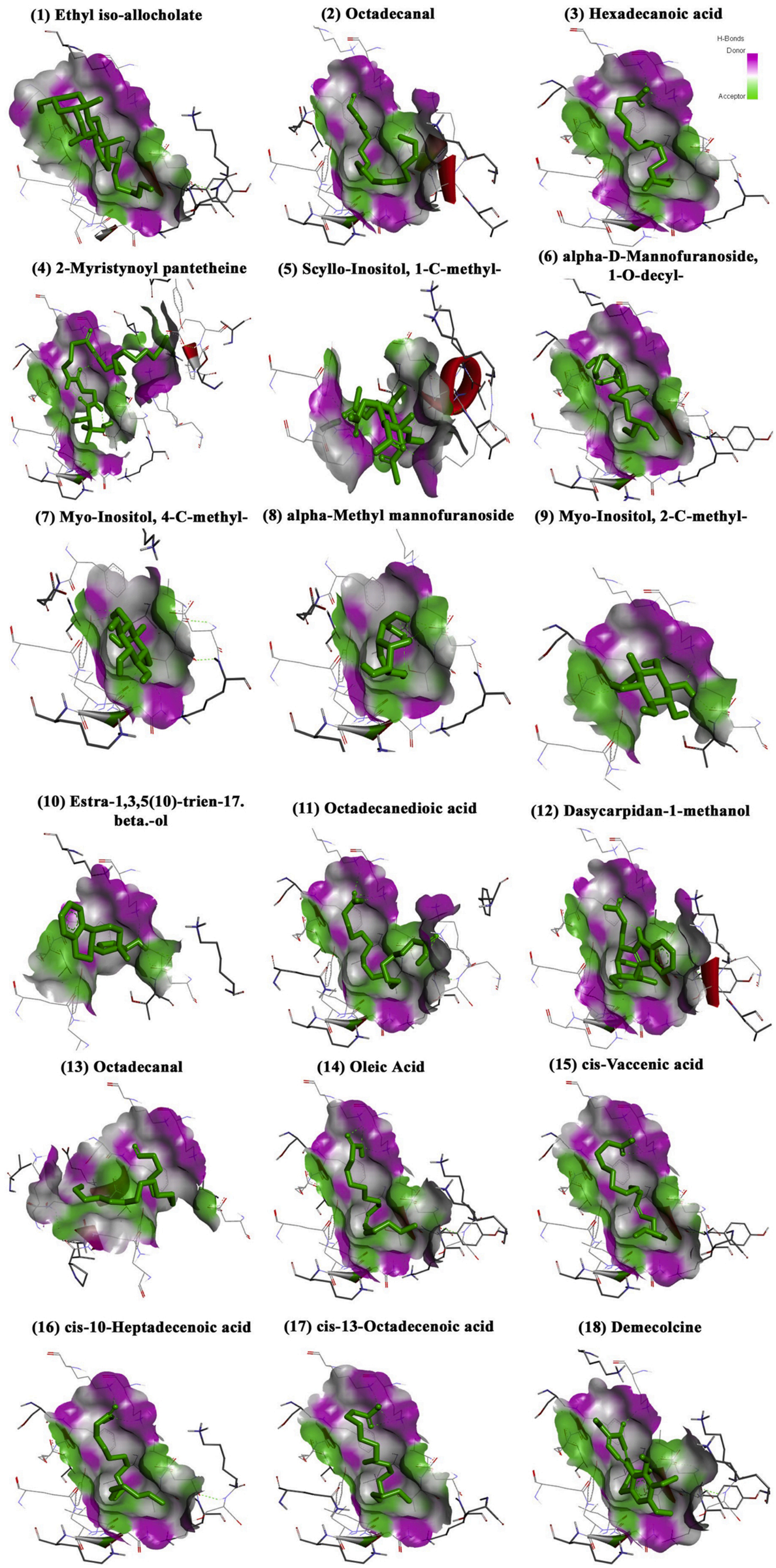

Fig. 6. Molecular docking complexes and interaction of the bioactive compounds of SMME with SarA.

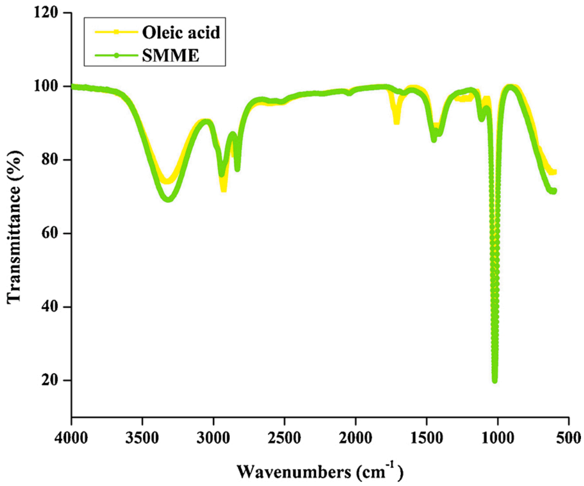

Fig. 7. FTIR spectra of standard oleic acid and SMME.

icaA and *icaD* are involved in the production of PIA and down regulation of these genes can be related with inhibition of slime synthesis by SMME (Fig. 4d). *fnbA*, *fnbB* and *cna* are adhesion proteins that interact with components of host extra cellular matrix and are involved in host colonization (Arciola et al., 2005; Haddad et al., 2018). Reduced expression of these genes is responsible for the antibiofilm activity of SMME. *atlA* is the autolysin responsible for eDNA synthesis and decreased expression of *atlA* is in line with the inhibition of autolysis and eDNA synthesis by SMME. On the whole, it could be observed that SMME treatment impaired the synthesis of adhesive elements such as slime, polysaccharides and eDNA. Decreased expression of these adhesion factors could be the underlying mechanism of action for antibiofilm activity of SMME.

3.3. GC–MS based profiling of bioactive metabolites of SMME and their molecular interaction with SarA of MRSA

S. mukorossi is well known for having plethora of medicinal properties and produces numerous beneficial secondary metabolites. The present study evaluated the strong antibiofilm potential of SMME. In order to partially extract the active principle(s) responsible for the antibiofilm potential, pericarp powder of *S. mukorossi* was subjected to successive solvents extraction using solvents. The result of successive solvents extraction showed that methanol fraction has maximum antibiofilm activity (73 %) when compared to the other solvent fractions (Fig. 5a). Hence, this methanol fraction was further subjected to GC–MS analysis (Fig. 5b) and the list of compounds present in methanol fraction is shown in Table 2. In order to identify the bioactive compound of SMME which is responsible for antibiofilm activity against MRSA, molecular docking analysis was performed against SarA protein of MRSA. *sarA* locus produces a nucleic acid-binding protein SarA (staphylococcal accessory regulatory A) which is a global regulator and controls the expression of various virulence factors and biofilm formation in *S. aureus* (Cheung et al., 2008; Snowden et al., 2013). The previous study

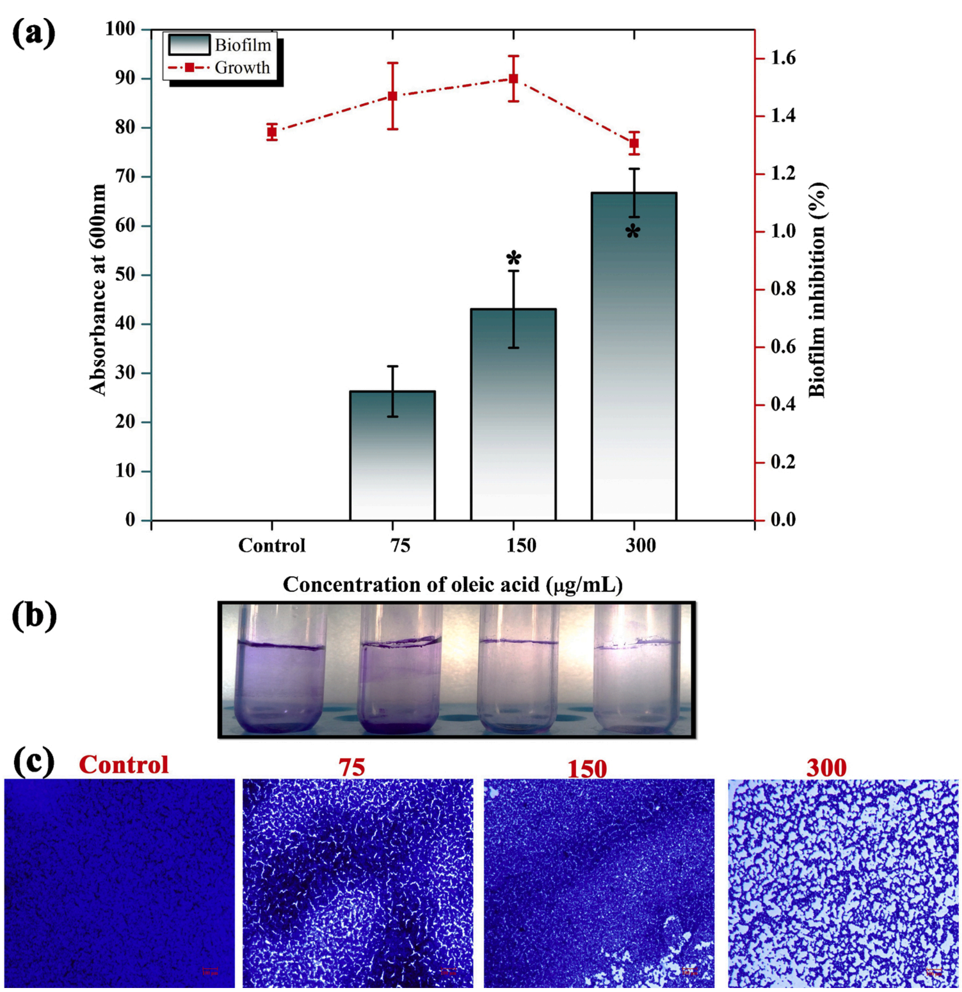

Fig. 8. Antibiofilm efficacy of oleic acid against MRSA (a). Effect of oleic acid on air-liquid interface MRSA biofilm (b). Inhibitory potential of oleic acid on biofilm formation of MRSA on glass surfaces (c). Error bars indicate standard deviations and asterisk indicates the statistical significance ($p < 0.05$).

demonstrated that the deletion of *sarA* locus diminished the ability of *S. aureus* to form biofilm (Beenken et al., 2003). In addition, compounds targeting SarA have already been reported to inhibit the biofilm formation in *S. aureus* (Arya and Princy, 2013; Balamurugan et al., 2017). Therefore, SarA was retrieved from PDB database and used as molecular target for bioactive compounds of SMME identified by GC–MS. The binding energy of bioactive compounds of SMME is listed in the Table 3 and the interactions are displayed in Fig. 6. The result of molecular docking analysis revealed the highest interaction of oleic acid with SarA when compared with other bioactive metabolites of SMME. Oleic acid strongly interacted with SarA with the binding energy of - 6.85 kcal/-mol, and showed the two hydrogen bonding (Glu B: 135; Lys B: 127) interaction. Then, FTIR was used to validate the presence of oleic acid in the SMME. The results of FTIR showed similarity between SMME and commercially procured oleic acid in the spectral regions 3000–2800 cm^{-1}, 1700–1500 cm^{-1} and 1200–900 cm^{-1} relevant to lipids, amide bonds of proteins and polysaccharides respectively (Fig. 7). This result confirmed the presence of oleic acid in methanolic extract of *S. mukorossi*. Oleic acid, a mono-unsaturated omega-nine fatty acid and abundantly found in various plants such as Olea europaea (olive oil), *Hippophae salicifolia* (sea buckthorn oil), *Glycine max* (soybean oil), *Zea mays* (corn oil), *Vitis vinifera* (grapeseed oil), *Elaeis guineensis* (palm oil) and *Carthamus tinctorius* (safflower oil) (Choi et al., 2010; Sales-Campos et al., 2013; Piccinin et al., 2019). It has been reported to be used for the treatment of various diseases such as adrenoleukodystrophy (Rizzo et al., 1986), sepsis (Gonçalves-de-Albuquerque et al., 2016), blood pressure (Terés et al., 2008) and coronary heart disease risk (Perdomo et al., 2015). It was also recognised for its antioxidant property (Emekli-Alturfan et al., 2010), antibacterial and antibiofilm activity against *S. aureus* (Stenz et al., 2008; Chen et al., 2011). Hence, Oleic acid was commercially procured and assessed for its antibiofilm activity by performing crystal violet based biofilm quantification assay, ring biofilm assay and light microscopic analysis. As predicted, oleic acid inhibited the biofilm formation of MRSA up to 67 % at 300 μg/mL (Fig. 8). Thus, the present study unveiled the antibiofilm activity of *S. mukorossi* and also identified oleic acid as the active molecule responsible for antibiofilm activity against MRSA. However, the antibiofilm potential of SMME is highly efficacious than oleic acid which presumed to the hypothesis that oleic acid could interact synergistically with other compounds present in SMME to exert potent antibiofilm activity. Thus, further in depth research will throw more light on biofilm inhibitory potential of *S. mukorossi* plant for which the present study will serve as the platform.

4. Conclusion

The present study evaluated the antibiofilm potential of SMME against clinically important pathogen MRSA. SMME exhibited strong antibiofilm activity against MRSA without affecting the growth. It was found to inhibit the adhesive molecules and biofilm-associated virulence factors of MRSA which was further validated through qPCR analysis. GC–MS analysis and molecular docking represented the oleic acid as the major constituent responsible for antibiofilm potential of SMME. For the first time, the present study revealed the antibiofilm based therapeutic value of *S. mukorossi* against MRSA. This pilot study suggests that *S. mukorossi* could be a promising antibiofilm agent against MRSA biofilm-related infections.

CRediT authorship contribution statement

Anthonymuthu Selvaraj: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing - original draft. **Alaguvel Valliammai:** Investigation, Writing - original draft. **Muruganatham Premika:** Investigation. **Arumugam Priya:** Investigation. **James Prabhanand Bhaskar:** Writing - review & editing, Supervision. **Venkateswaran Krishnan:** Writing - review & editing, Supervision. **Shunmugiah Karutha Pandian:** Conceptualization, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no competing interests.

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